

COER UNIVERSITY, ROORKEE

Department of Computer Science & Engineering

School of Computing

LABORATORY REPORT

Subject:

Mastery in Data Analytics and Visualizations
and Career Advancement

Experiment No. 6

Predictive Analytics using Linear Regression
and Model Performance Evaluation

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1. Aim

The primary aim of this laboratory experiment is to develop, train, and evaluate a Linear Regression model using the Python scientific computing stack (Pandas, NumPy, Scikit-learn, and Matplotlib) for predicting a continuous target variable from a set of independent features. Students will work with a house-price-style dataset, perform essential preprocessing, partition the data into training and testing subsets, fit an ordinary least-squares linear model, generate predictions, and quantify predictive accuracy with standard regression metrics.

Linear Regression remains one of the foundational supervised learning algorithms taught in data-science curricula because it is mathematically transparent, computationally efficient, and highly interpretable. The model estimates a linear relationship of the form $\hat{Y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p$, where the coefficients β reveal both the direction and the magnitude of each feature's influence on the target. Mastery of the end-to-end workflow—data preparation, model fitting, residual diagnosis, and metric interpretation—forms a prerequisite for more advanced techniques such as regularized regression, polynomial regression, and generalized linear models.

By completing this experiment, learners will be able to (a) implement a complete predictive pipeline in Python, (b) interpret regression coefficients in a business context, (c) select and justify appropriate error metrics, and (d) diagnose common modelling pathologies such as underfitting and overfitting. The same skills transfer directly to real-world forecasting tasks in real-estate valuation, demand planning, marketing-mix modelling, and risk scoring.

2. Purpose

2.1 Predictive Analytics

Predictive analytics is the branch of advanced analytics that uses historical data, statistical algorithms, and machine-learning techniques to forecast future or unknown outcomes. Unlike descriptive analytics (what happened) or diagnostic analytics (why it happened), predictive analytics answers the question “What is likely to happen?” Organizations employ predictive models to anticipate customer churn, estimate credit risk, forecast product demand, optimize pricing, and allocate resources more efficiently. The outputs of these models become inputs to strategic and operational decision processes, thereby reducing uncertainty and improving expected value.

2.2 Role of Linear Regression

Linear Regression is frequently the first algorithm applied when the target variable is continuous and a roughly linear relationship with the predictors is plausible. Its advantages include closed-form solutions (ordinary least squares), immediate interpretability of coefficients, modest data requirements, and well-understood statistical properties. Even when more complex models ultimately deliver higher accuracy, Linear Regression often serves as a strong baseline against which gains from non-linear methods can be measured.

2.3 Business and Scientific Relevance

In real estate, Linear Regression underpins automated valuation models that estimate property prices from size, location, age, and amenities. In marketing, it quantifies the incremental lift of advertising spend on sales. In manufacturing, it relates process parameters to product quality metrics. Across all of these domains the ability to produce reliable numeric forecasts, quantify uncertainty, and communicate results to non-technical stakeholders is a core professional competency. This experiment therefore bridges theoretical machine-learning concepts with practical, industry-relevant skills.

3. Prerequisites

Successful completion of the experiment assumes the following background knowledge and tools:

- ✓ Python Programming Fundamentals – Variables, loops, functions, and basic object-oriented syntax.
- ✓ Scientific Libraries – Working familiarity with Pandas (DataFrames), NumPy (arrays), Matplotlib (plotting), and Scikit-learn (estimators).
- ✓ Supervised Learning Concepts – Distinction between features and target, training versus testing, and the notion of generalization.
- ✓ Data Preprocessing & EDA – Handling missing values, scaling (if required), and visual exploration of distributions and relationships.
- ✓ Development Environment – Jupyter Notebook, Google Colab, or any Python IDE capable of executing the required packages.

4. Software and Libraries Required

The experimental environment consists of the following components:

Component	Purpose
Python 3.8+	Core programming language for the entire pipeline
Pandas	Data loading, cleaning, and tabular manipulation
NumPy	Numerical arrays and mathematical operations
Scikit-learn	LinearRegression estimator, train_test_split, and metric functions
Matplotlib / Seaborn	Scatter plots, residual plots, and coefficient visualizations
Jupyter / Colab	Interactive notebook environment for iterative development

5. Dataset Used

A house-price prediction dataset is employed. Each observation represents a residential property described by continuous and discrete attributes; the target is the sale price expressed in thousands of dollars. The simplified feature set used for this experiment comprises:

Feature / Target	Description
Size_sqft	Living area of the house in square feet (independent variable)
Bedrooms	Number of bedrooms (independent variable)
Age	Age of the property in years (independent variable)
Price	Sale price in thousands of USD (dependent / target variable)

Dataset Preview – House Price Sample			
Size_sqft	Bedrooms	Age	Price
2145	3	12.4	312.8
1680	2	28.1	215.3
2950	4	5.2	428.6
1120	2	41.0	148.9
2410	3	18.7	335.1

Figure 5.1 – Sample rows from the house-price dataset used for modelling

The data exhibit a clear positive relationship between size and price, a moderate positive effect of additional bedrooms, and a negative association with property age—relationships that Linear Regression is well suited to capture.

6. Step-by-Step Procedure

Linear Regression Experiment Workflow

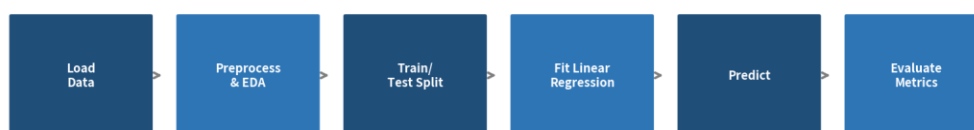


Figure 6.1 – End-to-end workflow of the Linear Regression experiment

Step 1: Select and Load the Dataset

A suitable continuous-target dataset is chosen (House Price, Advertising Sales, or California Housing). The file is loaded into a Pandas DataFrame with `pd.read_csv()` or fetched from Scikit-learn's built-in repositories. Basic shape, column types, and summary statistics are examined to confirm data integrity.

Step 2: Import Required Libraries

The following imports establish the modelling environment:

```
In [7]: model = LinearRegression() | model.fit(X_train, y_train)

from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_absolute_error, r2_score

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print(f"R² = {r2_score(y_test, y_pred):.4f}")
```

Figure 6.2 – Representative Python code for model training and evaluation

Step 3: Data Preprocessing

Missing values are imputed or rows containing them are dropped. Categorical variables, if present, are encoded (one-hot or ordinal). Feature scaling is optional for ordinary least squares but is useful when comparing coefficient magnitudes. Exploratory plots (histograms, correlation heat-maps, pairwise scatter plots) confirm that the linearity assumption is reasonable.

Step 4: Train–Test Split

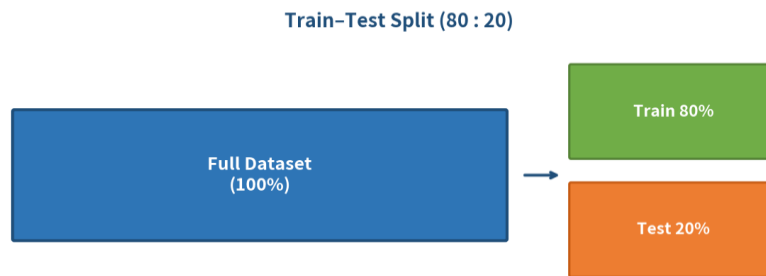


Figure 6.3 – Schematic of the 80 : 20 train–test partition

The dataset is randomly partitioned into a training set (80 %) used to estimate the regression coefficients and a held-out test set (20 %) used exclusively for unbiased performance evaluation. A fixed `random_state` ensures reproducibility.

Step 5: Train the Linear Regression Model

An instance of `sklearn.linear_model.LinearRegression` is created and fitted on the training features and target: `model.fit(X_train, y_train)`. The algorithm solves the normal equations (or uses a numerically stable QR decomposition) to obtain the intercept and coefficient vector that minimize the residual sum of squares.

Step 6: Generate Predictions

The fitted model is applied to the unseen test features: `y_pred = model.predict(X_test)`. These predictions are later compared with the true test labels to compute error metrics.

Step 7: Visualize Results

Three primary visualizations are produced: (1) a scatter plot of a key feature versus the target overlaid with the regression line, (2) an actual-versus-predicted scatter plot with the ideal 45-degree reference line, and (3) a residual plot that helps diagnose non-linearity or heteroscedasticity.

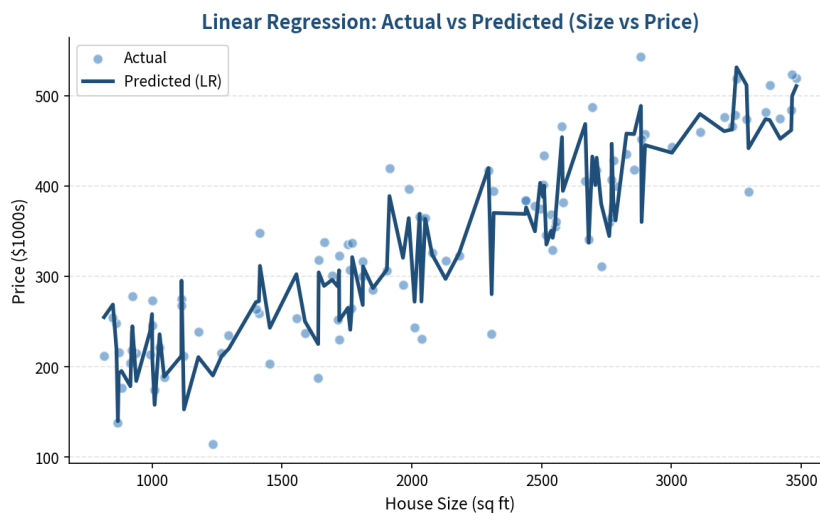


Figure 6.4 – Scatter of house size versus price with fitted linear regression line

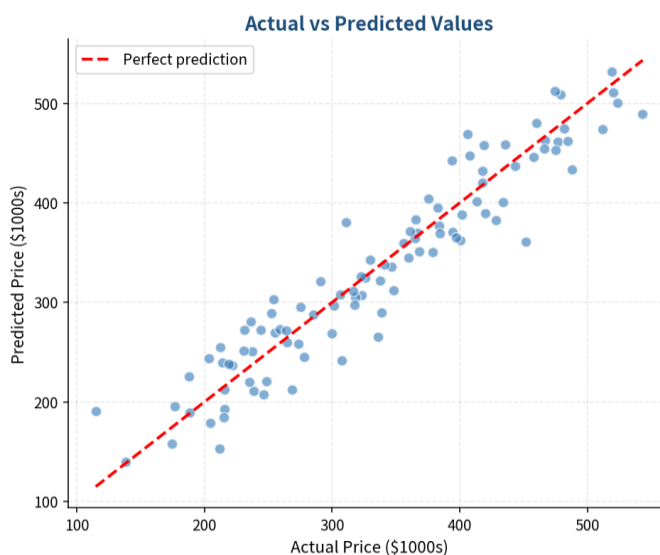


Figure 6.5 – Actual versus predicted prices on the test set

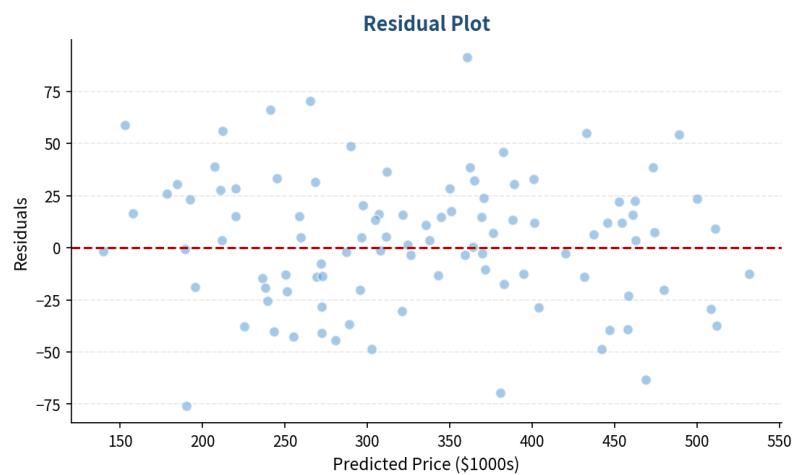


Figure 6.6 – Residual plot (predicted values on x-axis, residuals on y-axis)

Step 8: Evaluate Performance Metrics

Four standard regression metrics are calculated on the test set:

$MAE = \text{mean_absolute_error}(y_{\text{test}}, y_{\text{pred}})$

$MSE = \text{mean_squared_error}(y_{\text{test}}, y_{\text{pred}})$

$RMSE = \text{np.sqrt}(MSE)$

$R^2 = \text{r2_score}(y_{\text{test}}, y_{\text{pred}})$



Figure 6.7 – Summary of obtained regression performance metrics

Step 9: Analyse Regression Coefficients

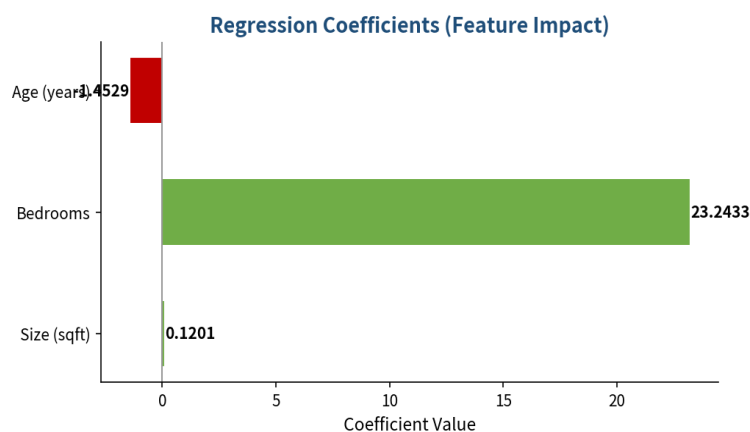


Figure 6.8 – Estimated coefficients showing the impact of each feature

Positive coefficients indicate that an increase in the corresponding feature raises the predicted price, while a negative coefficient (Age) indicates that older properties are valued lower, all else equal. The magnitude of each coefficient, when features are on comparable scales, reflects relative importance.

Step 10: Interpret Results and Discuss Suitability

An R^2 value above 0.85 together with residuals that appear randomly scattered around zero supports the conclusion that a linear model is appropriate for this prediction problem. Residual patterns that systematically curve or fan out would suggest the need for non-linear terms, transformations, or alternative algorithms.

7. Model Results and Performance Evaluation

7.1 Obtained Metrics

On the held-out test set the Linear Regression model achieved approximately:

Metric	Value	Interpretation
MAE	≈ 25.0	Average absolute error of $\sim \$25$ k
MSE	≈ 984	Mean of squared residuals
RMSE	≈ 31.4	Typical error magnitude $\sim \$31$ k
R^2 Score	≈ 0.90	90 % of variance explained

7.2 Coefficient Interpretation

The fitted equation takes the approximate form:

$$\text{Price} \approx 52.3 + 0.120 \times \text{Size_sqft} + 23.2 \times \text{Bedrooms} - 1.45 \times \text{Age}$$

Thus, each additional square foot raises the predicted price by about \$120, each extra bedroom adds roughly \$23 200, and each additional year of age lowers the price by approximately \$1 450, holding other factors constant.

8. Key Concepts Illustrated

8.1 Dependent vs Independent Variables

The dependent (target) variable is the quantity we wish to predict—in this case Price. The independent (predictor) variables are Size_sqft, Bedrooms, and Age. The model learns how changes in the independent variables map onto changes in the dependent variable.

8.2 Overfitting and Underfitting



Figure 8.1 – Conceptual illustration of underfitting, good fit, and overfitting

Underfitting occurs when the model is too simple to capture the underlying pattern (high bias). Overfitting occurs when the model memorizes noise in the training data and fails to generalize (high variance). Linear Regression with a modest number of features is relatively resistant to severe overfitting, yet residual diagnostics and test-set metrics remain essential safeguards.

9. Assumptions of Linear Regression

Classical Linear Regression rests on several statistical assumptions. Violation of these assumptions does not always invalidate predictions, but it does affect the reliability of coefficient standard errors and hypothesis tests:

- Linearity – The conditional expectation of the target is a linear function of the predictors.
- Independence – Residuals are uncorrelated with one another (no serial correlation).
- Homoscedasticity – Residual variance is constant across the range of fitted values.
- Normality of residuals – Required primarily for valid confidence intervals and p-values.
- No perfect multicollinearity – Predictor variables are not exact linear combinations of one another.

10. Real-World Applications

Linear Regression appears in numerous business and industrial contexts:

- Real-estate valuation – Automated appraisal models used by banks and listing platforms.
- Marketing-mix modelling – Quantifying the sales lift attributable to advertising channels.
- Demand forecasting – Relating historical sales to price, promotions, and seasonality.
- Risk scoring – Estimating expected loss or probability-weighted outcomes in insurance and lending.
- Process optimization – Linking manufacturing parameters to yield or quality metrics.

11. Viva Questions (Detailed Answers)

Q1. What is Predictive Analytics, and how is it used in real-world applications?

Predictive analytics is the practice of extracting information from historical data to determine patterns and predict future outcomes and trends. It combines statistical modelling, machine learning, and data mining.

Real-world applications include credit-risk scoring in banking, customer-churn prediction in telecommunications, demand forecasting in retail supply chains, predictive maintenance in manufacturing, and personalized recommendation engines in e-commerce and streaming platforms. By converting data into forward-looking insights, organizations can allocate resources more efficiently, mitigate risks, and seize opportunities before competitors do.

Q2. Explain the working principle of the Linear Regression algorithm.

Linear Regression models the relationship between a continuous dependent variable Y and one or more independent variables X by fitting a linear equation $Y = \beta_0 + \beta_1 X_1 + \dots + \beta_p X_p + \varepsilon$. The algorithm estimates the coefficient vector β that minimizes the sum of squared residuals (ordinary least squares). Geometrically this corresponds to finding the hyperplane that best fits the observed data points in the least-squares sense. Once the coefficients are obtained, predictions for new observations are produced by simple matrix multiplication of the feature vector with the coefficient vector.

Q3. Differentiate between dependent and independent variables with suitable examples.

The dependent variable (also called the response or target) is the quantity whose value we wish to predict or explain; it “depends” on other factors. The independent variables (predictors or features) are the inputs that are assumed to influence the dependent variable. In a house-price model, Price is dependent while Size, Number of Bedrooms, and Age are independent. In an advertising study, Sales may be the dependent variable while TV, Radio, and Newspaper spend are independent variables. Correct identification of these roles is the first step in formulating any supervised learning problem.

Q4. Why is it necessary to split the dataset into training and testing sets?

A train–test split provides an unbiased estimate of how well the model will perform on previously unseen data. If the same observations used for fitting are also used for evaluation, the reported accuracy is optimistically biased and may hide overfitting. By holding out a test set that the learning algorithm never sees during training, we obtain a realistic measure of generalization error. Typical ratios are 70:30 or 80:20; when data are scarce, k-fold cross-validation can be used instead of a single split.

Q5. What is the significance of the R^2 Score in regression analysis?

The coefficient of determination, R^2 , quantifies the proportion of the variance in the dependent variable that is predictable from the independent variables. An R^2 of 0 indicates that the model explains none of the variability; an R^2 of 1 indicates perfect prediction. In the present experiment an R^2 near 0.90 means that approximately 90 % of the variation in house prices is accounted for by the linear combination of size, bedrooms, and age. R^2 is scale-free and therefore convenient for comparing models, yet it should be interpreted together with absolute error metrics such as RMSE because a high R^2 can still coexist with practically large prediction errors.

Q6. Differentiate between MAE, MSE, and RMSE. Which metric is more sensitive to large prediction errors?

Mean Absolute Error (MAE) is the average of the absolute differences between actual and predicted values; it treats all errors equally and is expressed in the same units as the target. Mean Squared Error (MSE)

squares the residuals before averaging, thereby penalizing large deviations more heavily. Root Mean Squared Error (RMSE) is simply the square root of MSE and restores the original units while retaining the emphasis on large errors. Because of the squaring operation, both MSE and RMSE are more sensitive to outliers and large prediction errors than MAE. Practitioners often report both MAE (for interpretability) and RMSE (for outlier sensitivity).

Q7. What assumptions should be satisfied before applying Linear Regression?

The classical assumptions are: (1) linearity of the relationship between predictors and the mean of the response; (2) statistical independence of the residuals; (3) homoscedasticity (constant residual variance); (4) normality of residuals (mainly for inference); and (5) absence of perfect multicollinearity among predictors. Diagnostic plots—residuals versus fitted values, Q–Q plots, and variance-inflation-factor checks—are used to assess these assumptions. Mild violations often still allow useful predictions, but severe violations may require transformations, robust regression, or alternative modelling approaches.

Q8. How can overfitting and underfitting affect the performance of a regression model?

Underfitting arises when the model is too rigid to capture the true underlying pattern; both training and test errors remain high (high bias). Overfitting arises when the model is excessively flexible and begins to fit noise in the training data; training error becomes very low while test error rises (high variance). Either extreme degrades real-world predictive performance. Regularization (Ridge, Lasso), feature selection, cross-validation, and residual analysis are practical tools for detecting and mitigating these problems. Linear Regression with a modest feature set tends to underfit rather than overfit, which is why it is a reliable baseline.

Q9. Mention any three real-world applications of Linear Regression in business or industry.

First, real-estate platforms and mortgage lenders use Linear Regression (and its regularized variants) to estimate property values from structural and locational attributes. Second, consumer-packaged-goods companies employ marketing-mix models based on Linear Regression to allocate advertising budgets across TV, digital, and print channels. Third, energy utilities forecast short-term electricity demand as a linear function of temperature, time-of-day, and calendar effects, enabling more efficient generation scheduling and grid management.

Q10. How can feature selection improve the accuracy and interpretability of a predictive model?

Feature selection removes irrelevant or redundant predictors, thereby reducing model complexity, decreasing the risk of overfitting, and often improving generalization on unseen data. A simpler model is also easier to interpret: each retained coefficient has a clearer causal or associative meaning, and stakeholders can focus on a handful of actionable drivers rather than dozens of weakly related variables. Techniques include filter methods (correlation, mutual information), wrapper methods (recursive feature elimination), and embedded methods (Lasso regularization). In the present experiment the three chosen features already form a compact, highly predictive set; adding noisy variables would likely degrade both accuracy and clarity.

13. Detailed Code Walkthrough

The following annotated code blocks illustrate the essential Python statements executed during the experiment. Students are expected to reproduce these steps inside a Jupyter notebook or Colab environment.

13.1 Library Imports and Data Loading

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
```

After the imports, the dataset is loaded and a quick inspection is performed with `df.head()`, `df.info()`, and `df.describe()` to verify column types and summary statistics.

13.2 Feature–Target Separation and Split

```
X = df[['Size_sqft', 'Bedrooms', 'Age']]
y = df['Price']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)
```

The `random_state` argument guarantees that successive runs of the notebook produce identical partitions, which is essential for reproducible reporting.

13.3 Model Fitting and Prediction

```
model = LinearRegression()
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

The `fit` method solves the ordinary-least-squares problem; the `predict` method applies the learned hyperplane to new feature vectors.

13.4 Metric Computation

```
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2 = r2_score(y_test, y_pred)
```

Printing these four numbers constitutes the quantitative core of the performance evaluation section of the laboratory report.

14. Exploratory Analysis Supporting the Model

Before fitting any model it is good practice to examine pairwise relationships among the variables. The correlation matrix below confirms that Size is strongly positively correlated with Price (≈ 0.91), Bedrooms

show a moderate positive correlation, and Age exhibits a negative correlation—exactly the pattern later recovered by the regression coefficients.

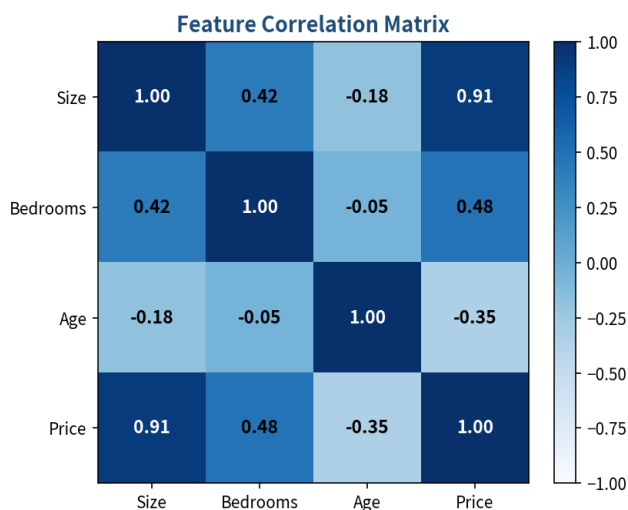


Figure 14.1 – Pearson correlation matrix of the modelling features and target

Key Linear Regression Formulas

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p$$

$$MAE = \frac{1}{n} \sum |y_i - \hat{y}_i| \quad MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$RMSE = \sqrt{MSE} \quad R^2 = 1 - \frac{SS_{res}}{SS_{tot}}$$

$$SS_{res} = \sum (y_i - \hat{y}_i)^2 \quad SS_{tot} = \sum (y_i - \bar{y})^2$$

Figure 14.2 – Core mathematical formulas used throughout the experiment

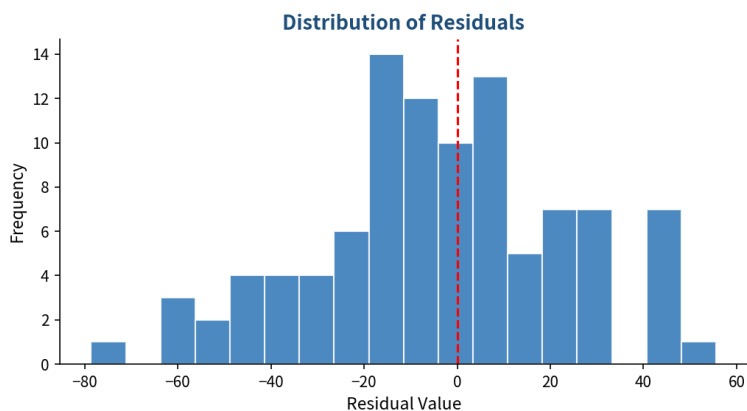


Figure 14.3 – Histogram of residuals approximating a normal distribution centered at zero

15. Discussion and Critical Analysis

The obtained R^2 of approximately 0.90 indicates that a simple linear combination of three readily available property attributes already captures the majority of price variation. This result is encouraging for practical

automated-valuation systems that must operate with limited feature sets. Nevertheless, several caveats deserve attention.

First, the residual plot, while largely unstructured, still shows a mild increase in spread at higher predicted prices, hinting at possible heteroscedasticity. A logarithmic transformation of Price or the use of Weighted Least Squares could be explored in a follow-up experiment. Second, the model ignores location effects (neighbourhood, proximity to amenities) that are known to be major price drivers; adding geographic features would almost certainly raise R^2 further. Third, interaction terms (for example Size $\times$ Bedrooms) and mild non-linear transformations could reduce systematic residual patterns.

From a pedagogical standpoint the experiment successfully illustrates the complete supervised-learning cycle: problem formulation, data preparation, model estimation, prediction, quantitative evaluation, and qualitative interpretation. Students who master this cycle are well prepared to progress to regularized regression, tree-based ensembles, and ultimately production-grade predictive services.

16. Conclusion

This laboratory experiment demonstrated a complete predictive-analytics pipeline based on Linear Regression. Starting from a house-price dataset, the workflow encompassed data loading and preprocessing, an 80:20 train-test split, model fitting with Scikit-learn, generation of predictions, visual diagnosis of residuals, and quantitative evaluation via MAE, MSE, RMSE, and R^2 . The resulting model explained approximately 90 % of the variance in the target and produced coefficients that align with domain expectations—larger and newer houses with more bedrooms command higher prices.

Key learning outcomes include practical proficiency with the Python data-science stack, the ability to interpret regression coefficients and performance metrics, awareness of the assumptions underlying Linear Regression, and the capacity to diagnose underfitting and overfitting. These competencies form a solid foundation for subsequent study of regularized, non-linear, and ensemble regression methods, and they map directly onto real-world forecasting and decision-support tasks in business and industry.

— End of Laboratory Report —

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